Effectiveness of Temporal Control Policy of Pedestrians for Congestion

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Introduction

In this paper we propose a framework to evaluate temporal control policy of the flow of a large number of pedestrians, and we show results of the evaluation by exhaustive simulation of pedestrians under various settings as case study.

When a disaster occurs or a big event is held, heavy congestion caused by simultaneous behaviors of a large number of people bring negative affects not only waste of time but also danger especially on elderly people and children. In order to solve this problem, we introduce an aspect of temporal view in addition to traditional spacial controls for pedestrians such as barricades. We conduct the evaluate experiment with computer simulation using simulator CrowdWalk.

Method

The proposed framework to evaluate the effectiveness of the various policies consists of exhaustive simulation rely upon temporal control policy. In order to evaluate the changes brought by the policies, such as pedestrian's satisfactions, we measured the time to destinations, walking speed and so on.

Temporal control : In the crowd moving such as an evacuation from a disaster, the time to start to move and the number of people are various. We assume that we can control them, and show how they affect the result.

Kamakura City where tsunami damage caused by earthquake is expected, is used as the target are of this case study.

Simulation

The simurater CrowdWalk adopts one-dimensional space model.When we input the place and the time of agent emerge and so on, CrowdWalk outputs the time took each agents or all agents to the destination.

Result

As the result of this reserch we can confirm that there is a possibility that we can decrease the congestion if we can control the crowd temporarily. But it has also the risk that evacuation is delayed if time adjustment is failed.

Fig.1 is one of the result of simulations . This figure shows that the evacuation time becomes long if the time we begin to evacuate is late, of course. But we notice that the ratio of agent emergence does not affect on evacuation time.

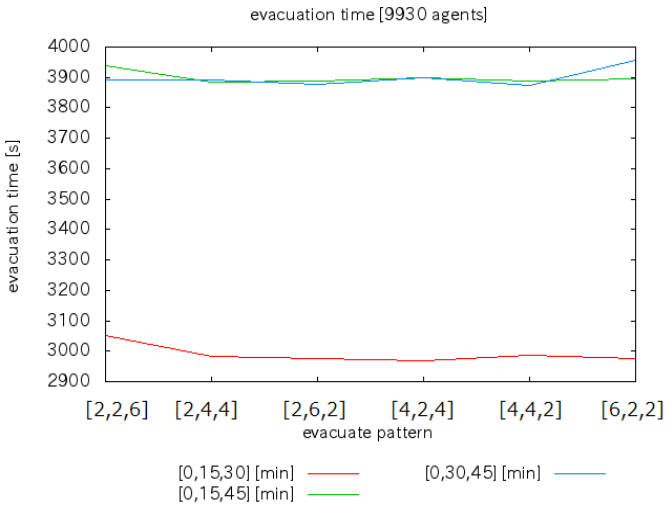


Fig.1 evacuation time

Fig.2 shows the integral of agents involved in the congestion. Here we can see the effect of the ratio of agent emergence. In the pattern where many agents emerge simultaneously, the congestion become worse.

In order to alleviate congestion, it is important to divide people properly.

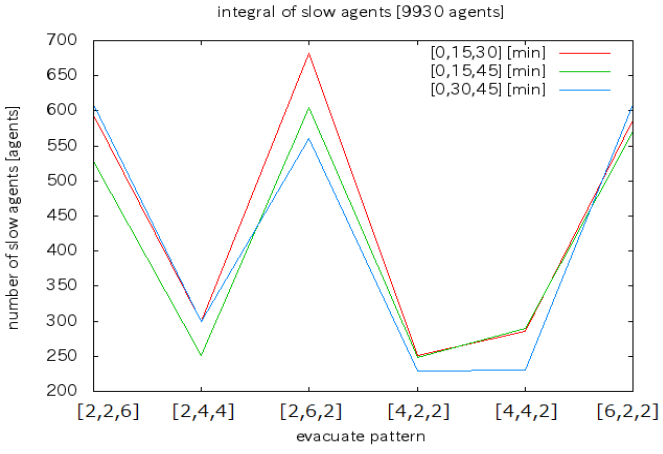


Fig.2 integral of slow agents

References

[1] Tomohisa YAMASHITA et al, Implementation of Simulation Environment for Exhaustive Analysis of Huge-Scale Pedestrian Flow SICE Journal of Control, Measurement , and System Integration,Vol.6,No.2,pp.137-146,March 2013